

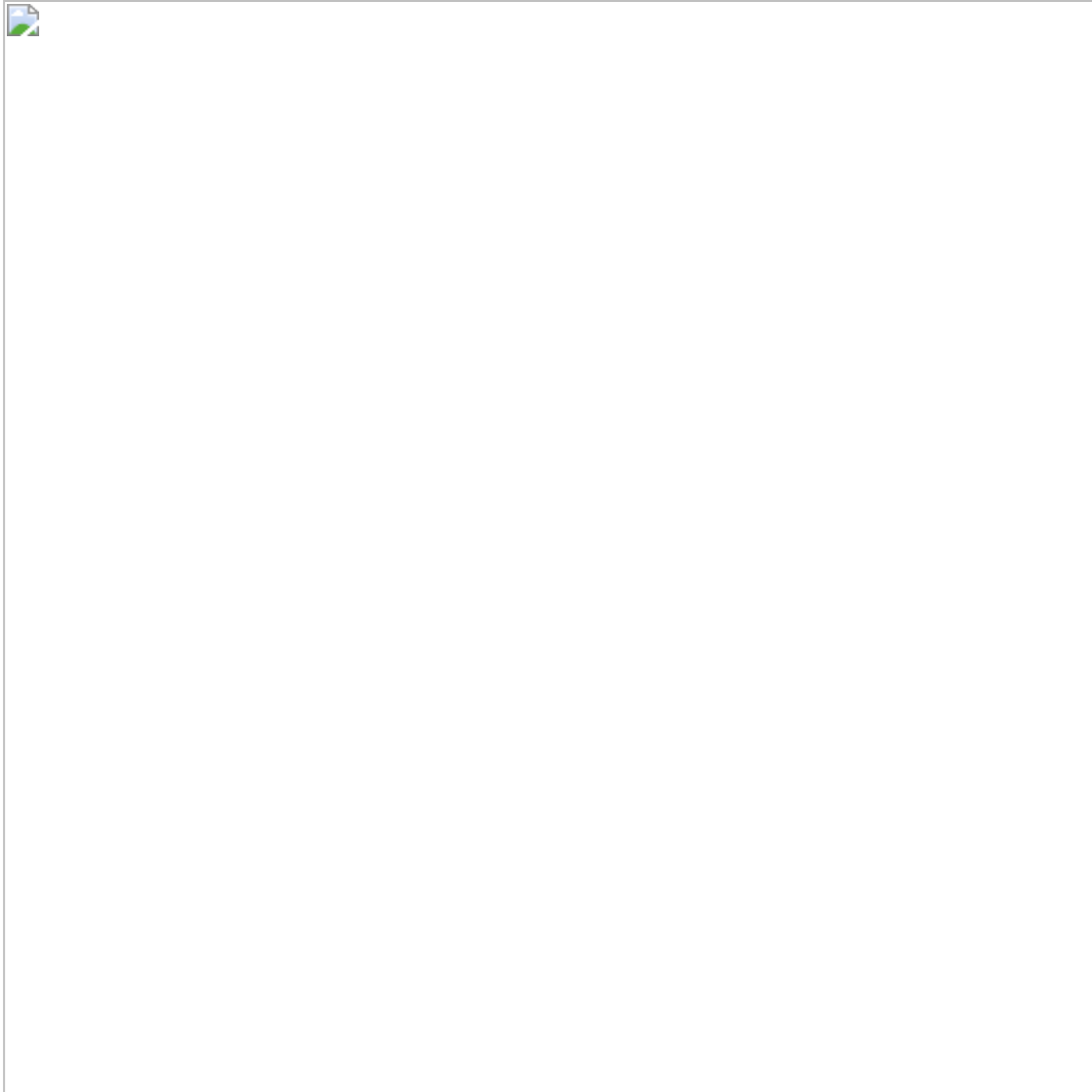
Antimicrobial Resistance Prediction from MALDI-TOF

[k kaggle.com/competitions/antimicrobial-resistance-prediction-from-maldi-tof/overview](https://kaggle.com/competitions/antimicrobial-resistance-prediction-from-maldi-tof/overview)

The image shows the Kaggle logo, which consists of the word "kaggle" in a light blue, lowercase, sans-serif font. A small "TM" trademark symbol is located at the top right of the letter "e". The logo is centered on a dark gray rectangular background.

Vanessa · Community Prediction Competition · Private · a day to go

Antimicrobial Resistance Prediction from MALDI-TOF Mass Spectrometry for students of the Master in Machine Learning for Health



Overview

Welcome to the Antimicrobial Resistance Prediction from MALDI-TOF Mass Spectrometry challenge!

In this competition, you will develop machine learning models to predict antimicrobial resistance (AMR) patterns from MALDI-TOF mass spectrometry data. The goal is to explore semi-supervised and multi-label learning strategies, combining labeled and unlabeled data to make accurate predictions across multiple antibiotics.

This challenge simulates a real-world biomedical ML problem, where:

- Only part of the data is labeled,
- The target involves multiple correlated outputs,
- Domain knowledge (microbiology) can guide the modeling process.

Start

2 months ago

Close

a day to go

Description

Scientific Context

MALDI-TOF MS is a rapid diagnostic tool widely used in clinical microbiology for bacterial identification and, increasingly, resistance prediction.

Step	Description	Approx. Time
1 Sample collection	Clinical sample obtained (urine, stool, blood, etc.)	—
2 Bacterial culture	Sample plated and incubated until isolated colonies grow	18–24 h
3 MALDI sample preparation	A colony is placed on the MALDI target and covered with matrix	5–10 min
4 MALDI-TOF analysis	Laser ionization and time-of-flight mass spectrometry acquisition	< 1 min

Predicting AMR directly from these spectra could:

- Reduce diagnostic time (minutes instead of days)
- Support optimal antibiotic therapy
- Contribute to antimicrobial stewardship efforts

Task

Predict, for each test sample, the resistance (0 or 1) to the 8 antibiotics.

This is a multi-label and semi-supervised classification challenge:

- Some training samples are fully labeled, others partially labeled
- You are encouraged to leverage unlabeled samples using semi-supervised techniques (e.g., pseudo-labeling, co-training, consistency regularization)

Besides, there are species-specific patterns, i.e. , resistance mechanisms vary across bacteria.

Evaluation

Kaggle Leaderboard Metric

Submissions are evaluated using the average AUC (Area Under the ROC Curve) across the 8 antibiotic prediction tasks.

For each antibiotic a_i :

—

$$AUC_{mean} = \frac{1}{8} \sum_{i=1}^8 AUC(a_i)$$

The public leaderboard is based on a random 40% of the test set. The private leaderboard (used for final grading) uses the remaining 60%.

Final Challenge Evaluation (Course Grading)

This Kaggle challenge represents 15% of the final course grade. The grading scale is based on the private leaderboard AUC following the next scale:

Metric	Grade
AUC = 0.50 (sample submission)	Score = 0
AUC = 0.80 (benchmark)	Score = 4
AUC = Best team's AUC	Score = 10

Intermediate AUC values will receive a linearly interpolated score between the two closest reference points. For example:

- If your AUC is between 0.50 and 0.80, your grade will range linearly between 0 and 4.
- If your AUC is between 0.80 and the best team's AUC, your grade will range linearly between 4 and 10.

An additional 10% of the grade will be assigned based on the poster presentation session.

Competition Host

Vanessa

Prizes & Awards

Kudos

Does not award Points or Medals

Participation

39 Entrants

37 Participants

11 Teams

380 Submissions

Tags

Mean Columnwise Area Under Receiver Operating Characteristic Curve

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